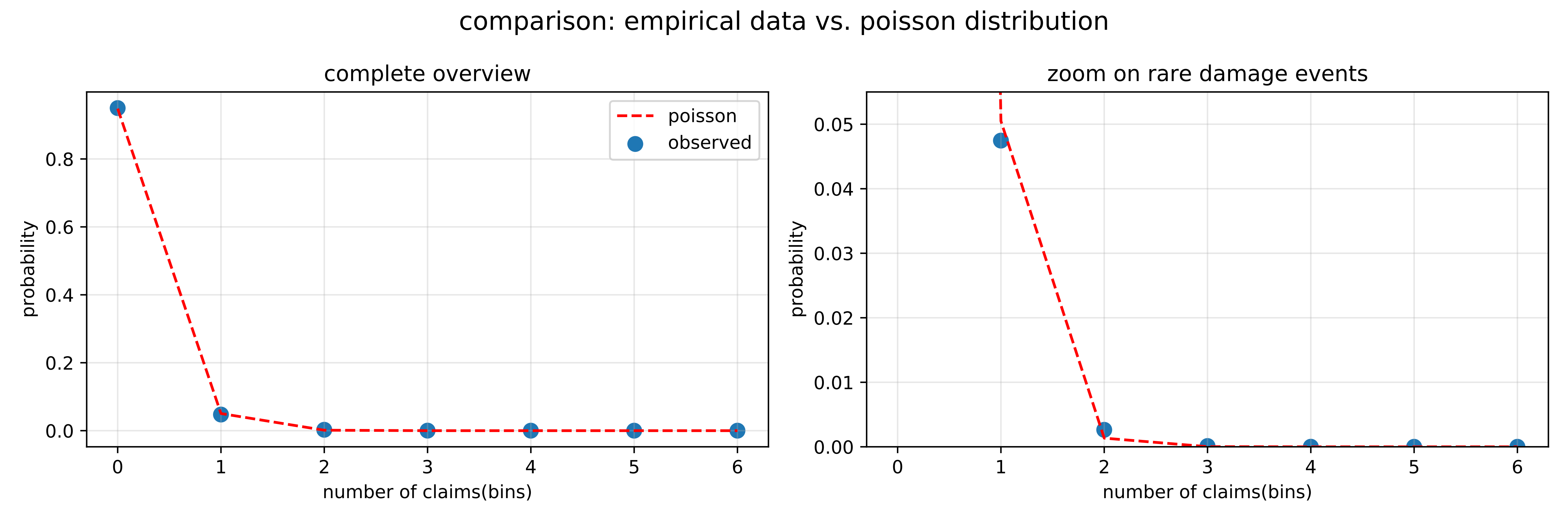
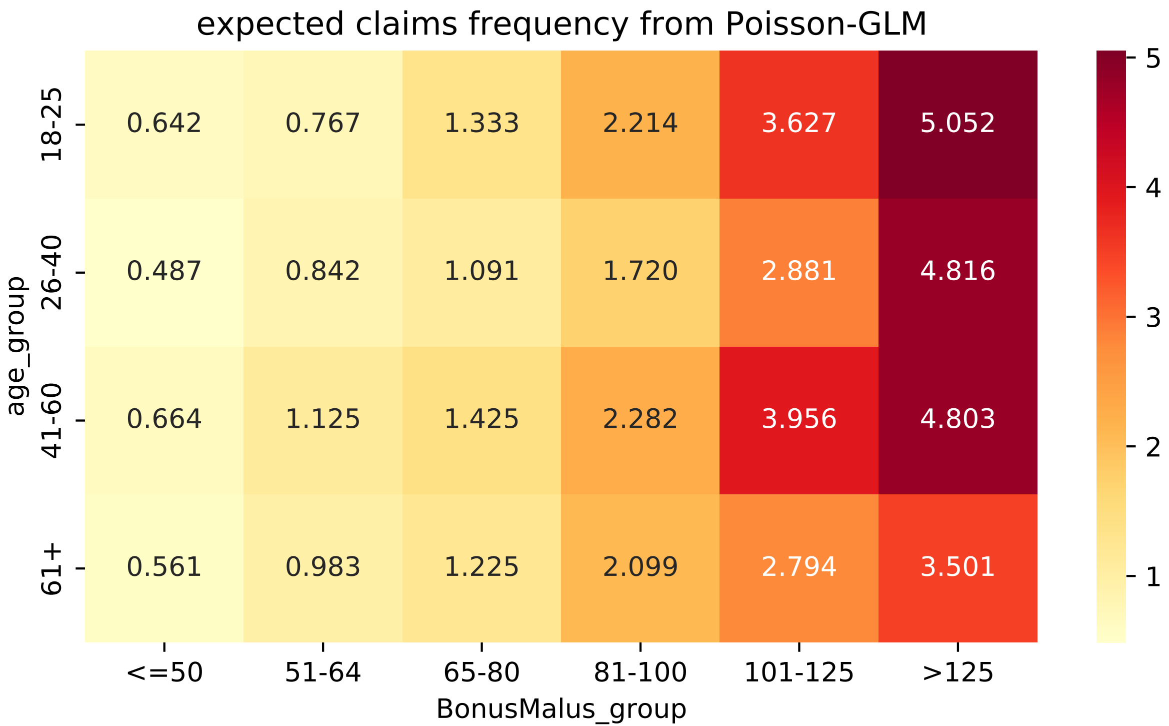
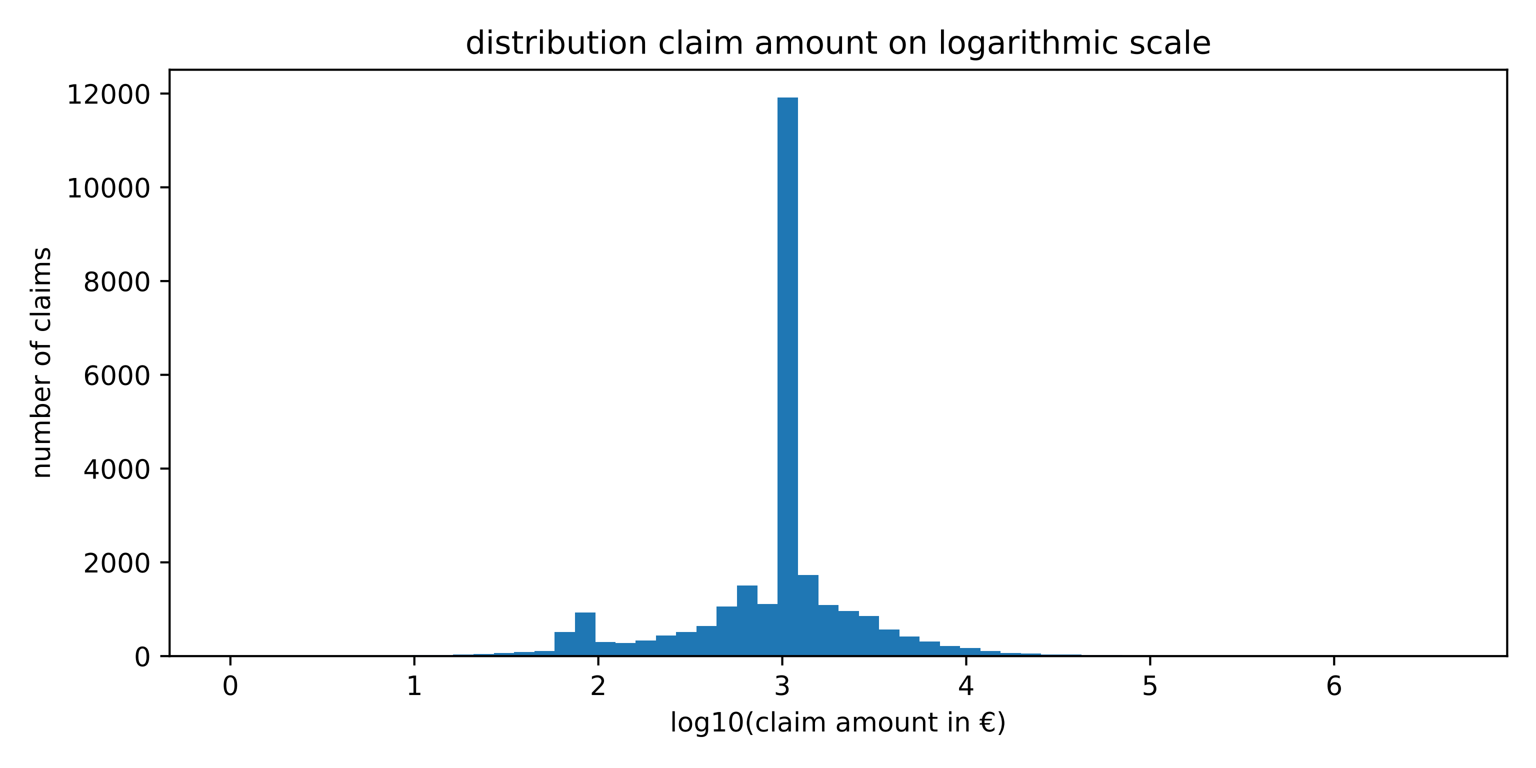
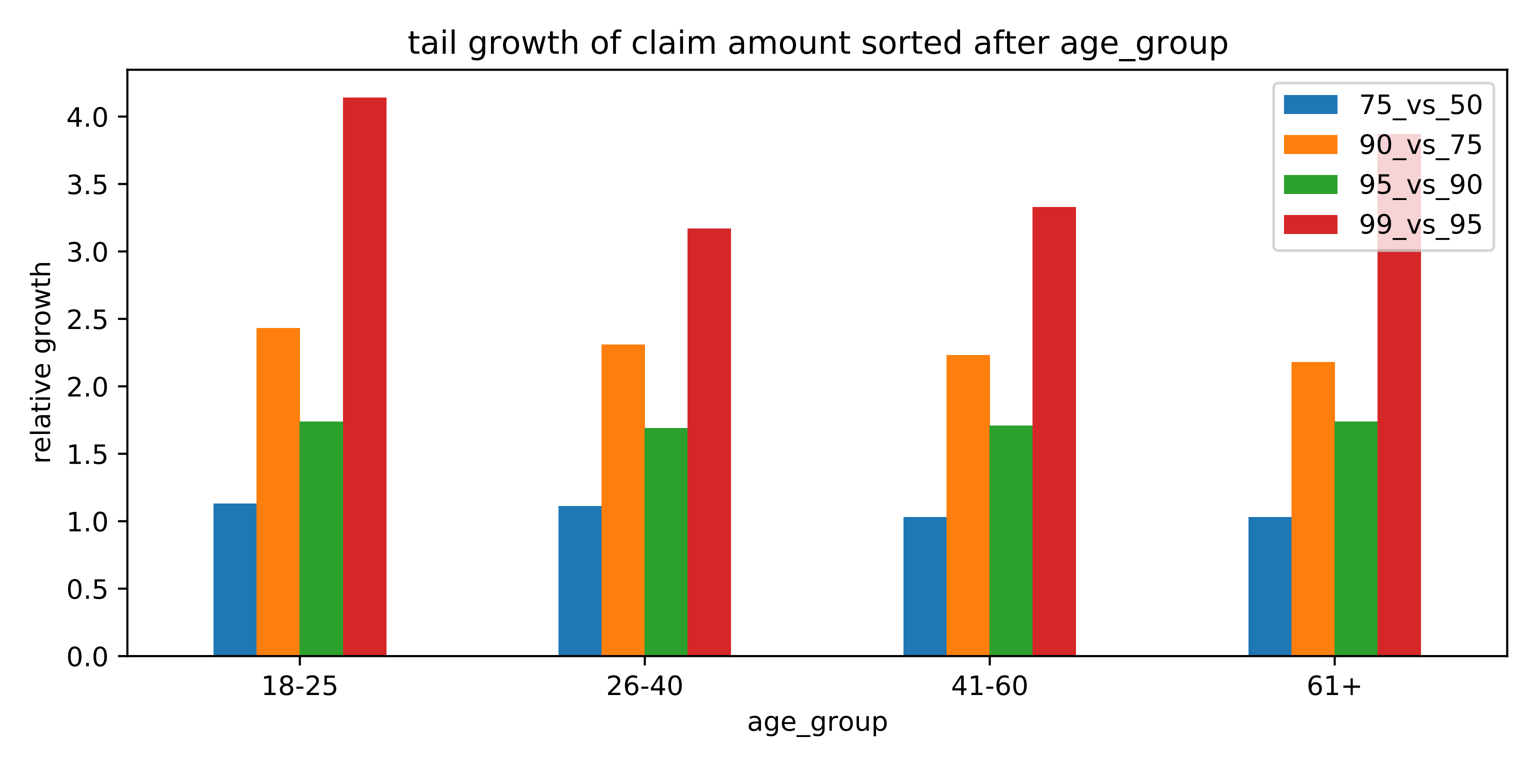


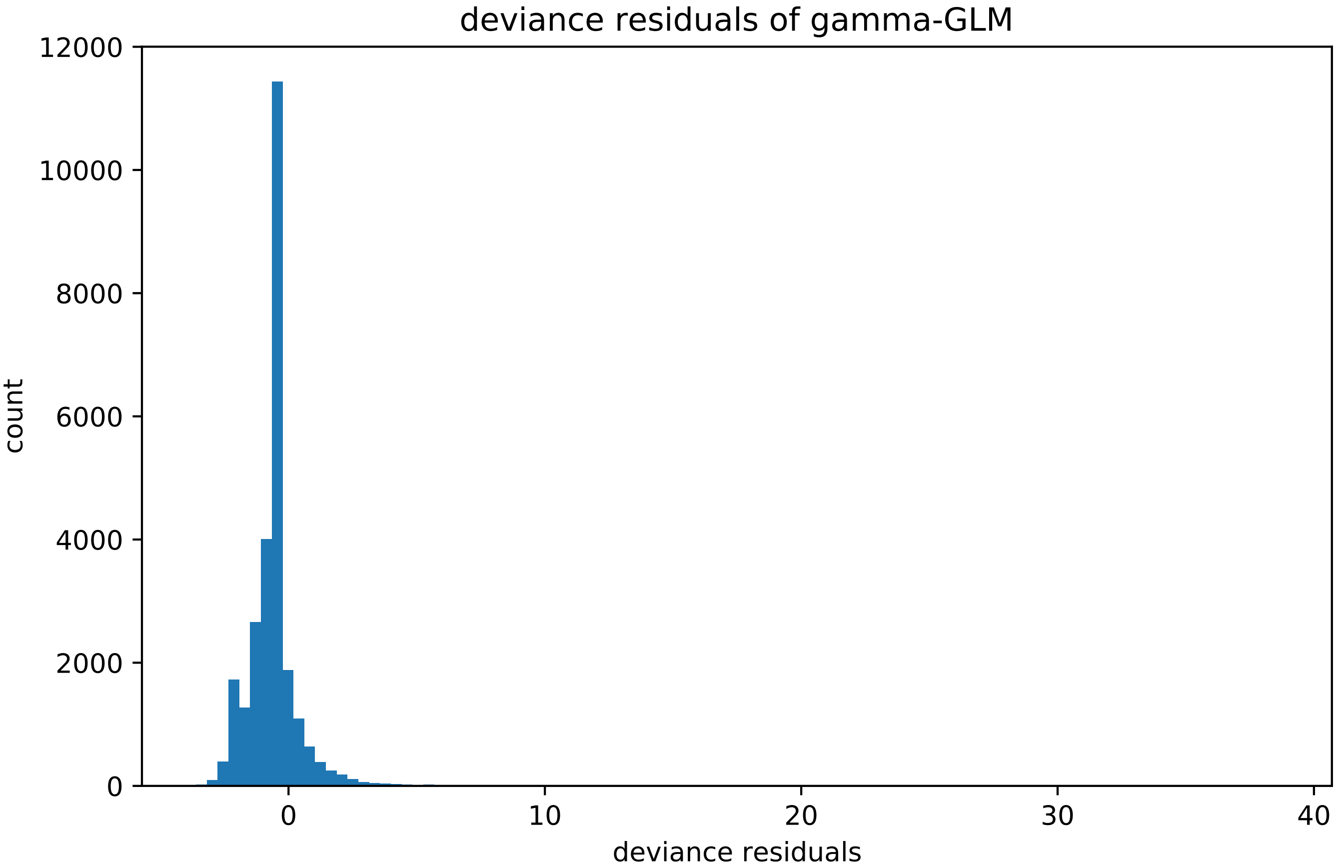
comparison: empirical data vs. poisson distribution
complete overview
zoom on rare damage events
poisson
observed
probability
number of claims(bins)
0.0
0.2
0.4
0.6
0.8
0.00
0.01
0.02
0.03
0.04
0.05
0
1
2
3
4
5
6

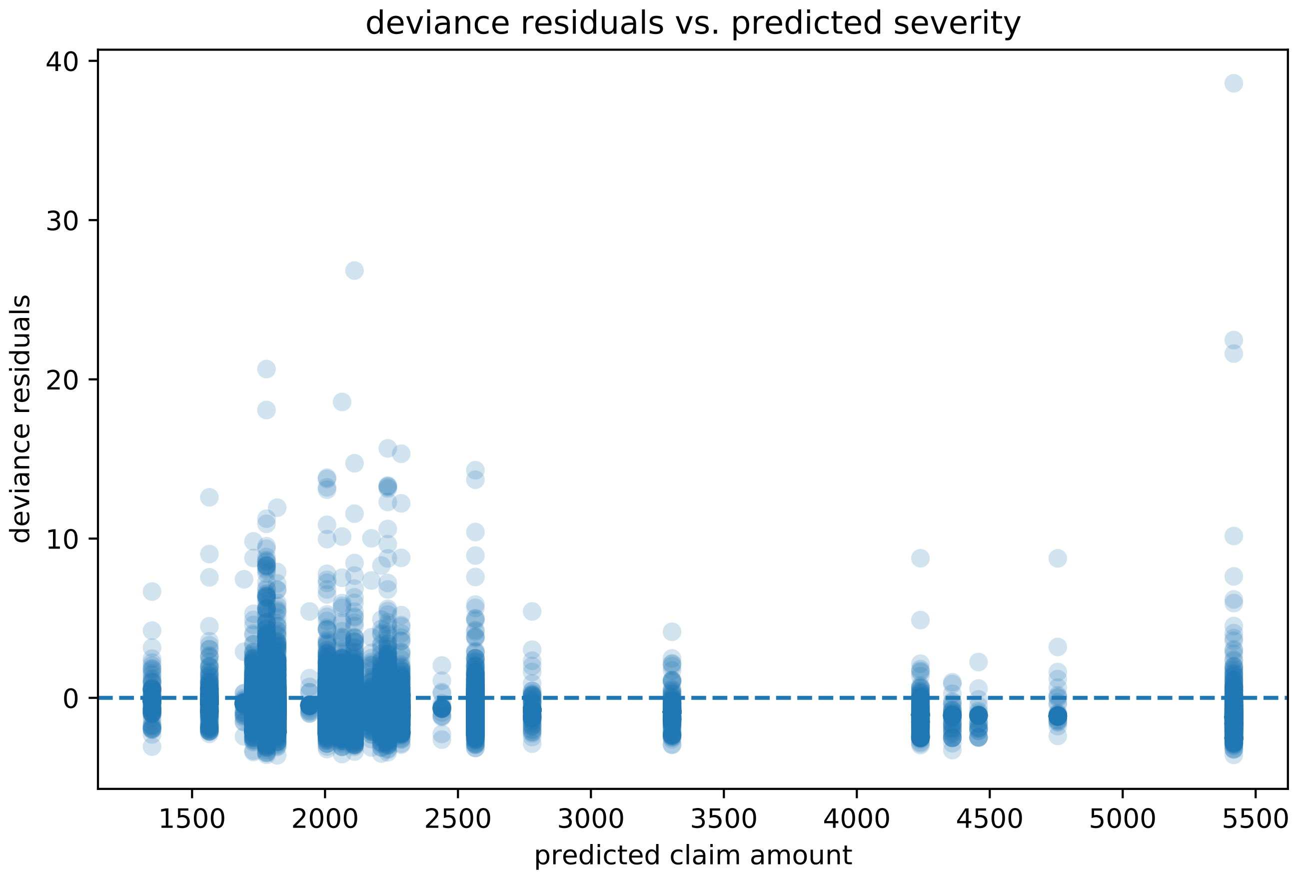

expected claims frequency from Poisson-GLM
age_group
18-25
26-40
41-60
61+
BonusMalus_group
<=50
51-64
65-80
81-100
101-125
>125
0.642
0.767
1.333
2.214
3.627
5.052
0.487
0.842
1.091
1.720
2.881
4.816
0.664
1.125
1.425
2.282
3.956
4.803
0.561
0.983
1.225
2.099
2.794
3.501
5
4
3
2
1

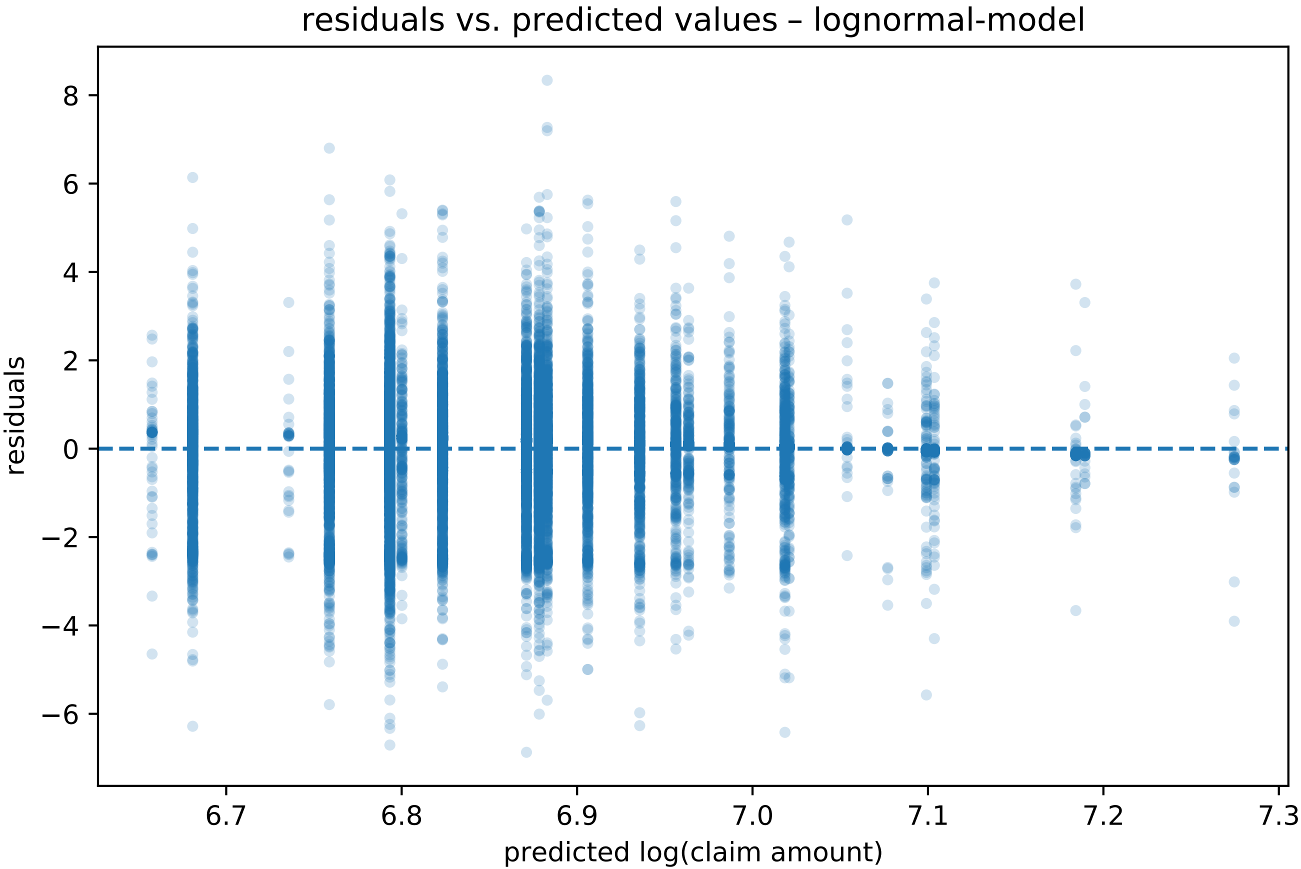

distribution claim amount on logarithmic scale
number of claims
0
2000
4000
6000
8000
10000
12000
0
1
2
3
4
5
6
log10(claim amount in €)

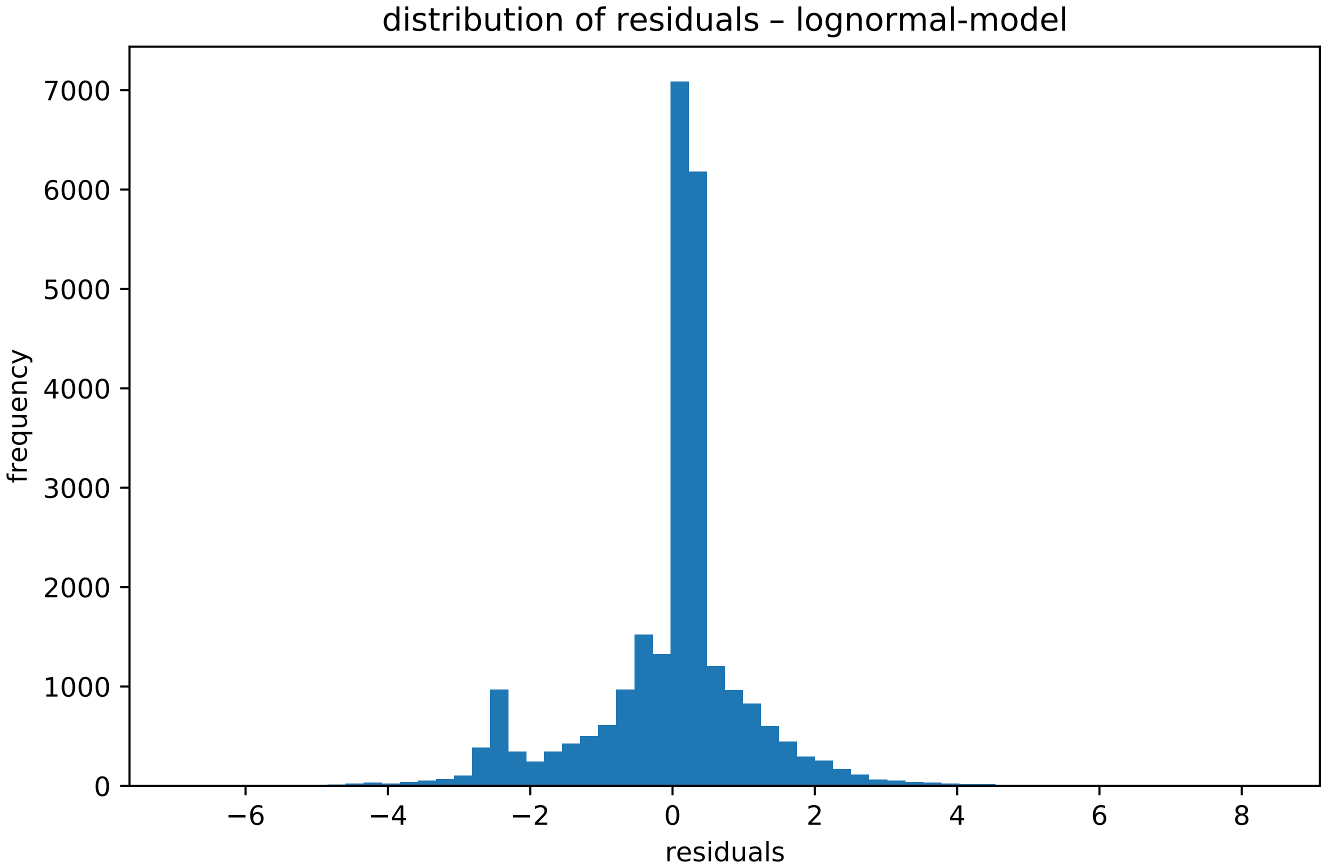


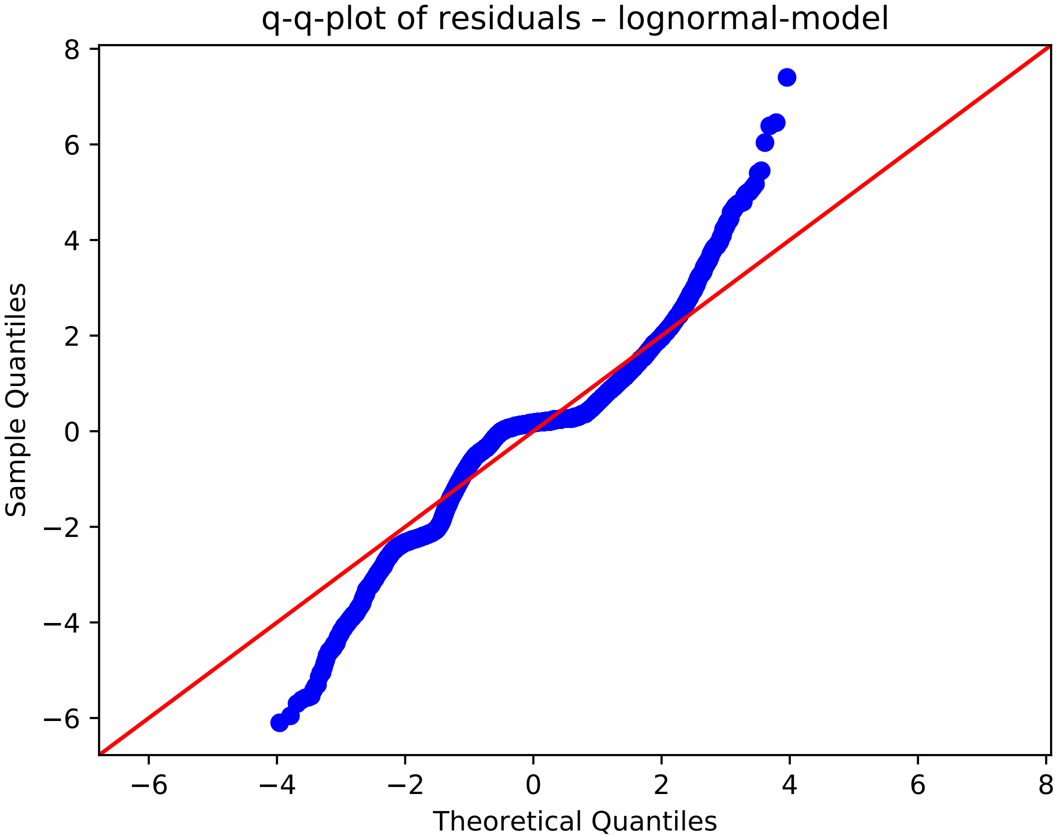
tail growth of claim amount sorted after age_group
relative growth
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
18-25
26-40
41-60
61+
age_group
75_vs_50
90_vs_75
95_vs_90
99_vs_95

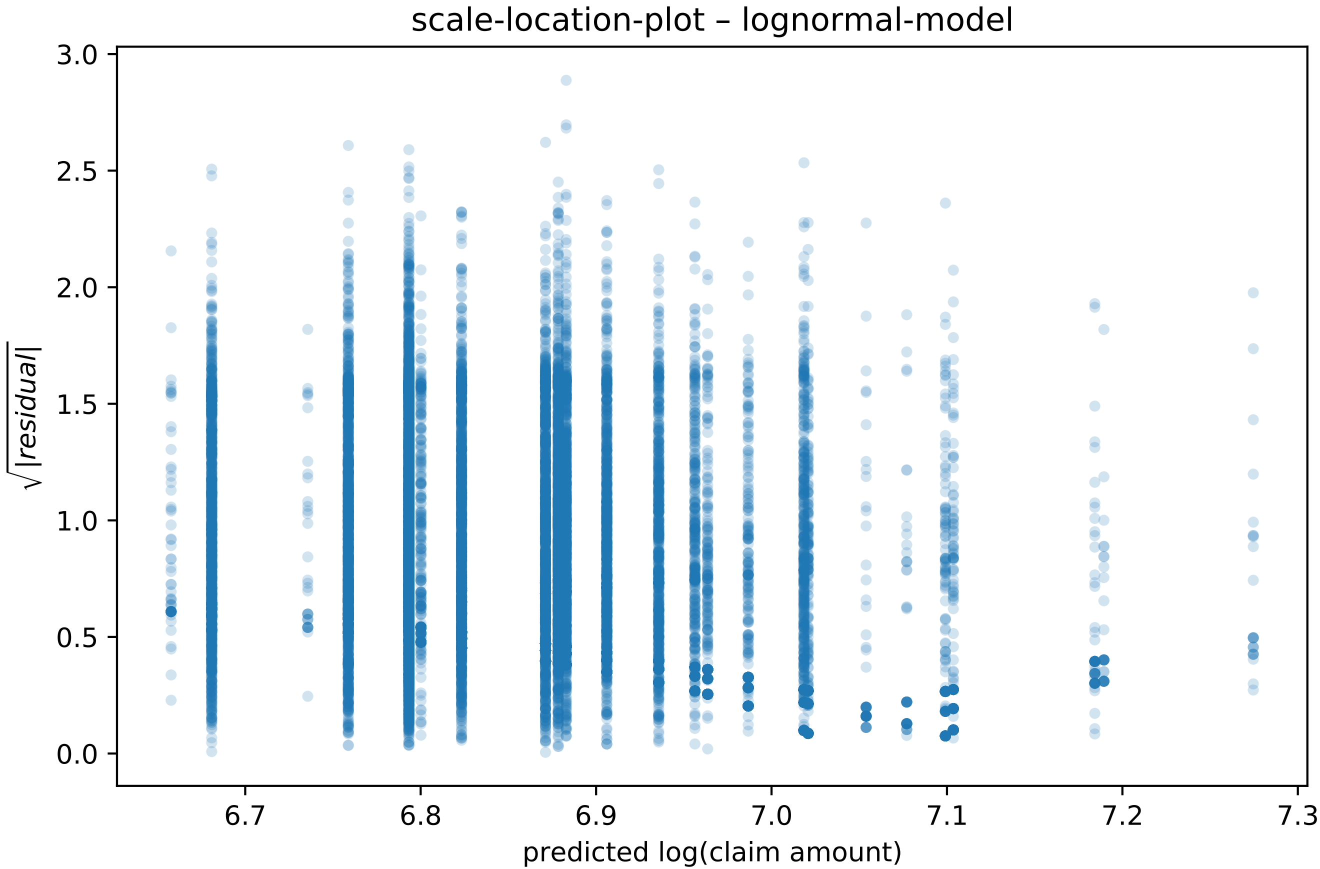

deviance residuals of gamma-GLM
count
12000
10000
8000
6000
4000
2000
0
0
10
20
30
40
deviance residuals


deviance residuals vs. predicted severity
deviance residuals
predicted claim amount
40
30
20
10
0
1500
2000
2500
3000
3500
4000
4500
5000
5500


residuals vs. predicted values – lognormal-model
8
6
4
2
0
−2
−4
−6
residuals
6.7
6.8
6.9
7.0
7.1
7.2
7.3
predicted log(claim amount)


distribution of residuals – lognormal-model
7000
6000
5000
4000
3000
2000
1000
0
frequency
−6
−4
−2
0
2
4
6
8
residuals


q-q-plot of residuals – lognormal-model
Sample Quantiles
Theoretical Quantiles
8
6
4
2
0
−2
−4
−6
−6
−4
−2
0
2
4
6
8


scale-location-plot – lognormal-model
$\sqrt{|residual|}$
predicted log(claim amount)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
6.7
6.8
6.9
7.0
7.1
7.2
7.3

model comparison by age_group

	observed_mean	gamma_mean	lognormal_mean	count
18-25	5121.62	4826.9	1842.49	2403.0
26-40	2115.68	2139.58	1702.28	8179.0
41-60	1797.83	1805.61	1777.12	11893.0
61+	2246.47	2255.38	1923.5	3969.0

$$WMRAE = \sum w_g \left|\frac{y_g - y_o}{y_g}\right|$$

WMRAE (Age Groups) = 1.13%

model comparison by bm_group

	observed_mean	gamma_mean	lognormal_mean	count
<=50	1922.22	1935.55	1687.83	11592.0
51-64	2022.05	2025.75	1757.43	5516.0
65-80	2039.67	2124.38	1829.31	3974.0
81-100	3849.88	3544.64	1928.76	4106.0
101-125	1844.08	2209.53	2094.51	1093.0
>125	3339.58	3078.19	2345.35	163.0

$$WMRAE = \sum w_g \left|\frac{y_g - y_o}{y_g}\right|$$

WMRAE (Bonus-Malus Groups) = 3.06%